9.

a) First (S) = First (A-ε) ∪ First (a) ∪ First (b) ∪ First (B-ε) ∪ First(d)

= d, a, b

First (A) = d, ε

First (B) = d, ε

b) Follow (S) = $

Follow (A) = First (a) ∪ First (c)

= a, c

Follow (B) = First (d) = d

c)

	a	b	c	d	
S	S→Aa	S→bAc		S→Bd	
A	A→ε		A→ε	A→d	
B				B→dc	

∵ First (B) and Follow (B) both contain d,

∴ A conflict exists and thus the grammar not LL(1).

Question 9 [30%]
Consider the following grammar:

1. S → Aa | bAc | Bd
2. A → d | ε
3. B → dc | ε

In this grammar:
ε represents NULL;
Non-Terminals: S, A, B;
Terminals: a, b, c, d, ε;

(a) Calculate the First sets for each non-terminal in the grammar. [10%]
(b) Calculate the Follow sets for each non-terminal in the grammar. [10%]
(c) Construct the LL(1) parsing table for the grammar and determine if the grammar is LL(1). [10%]

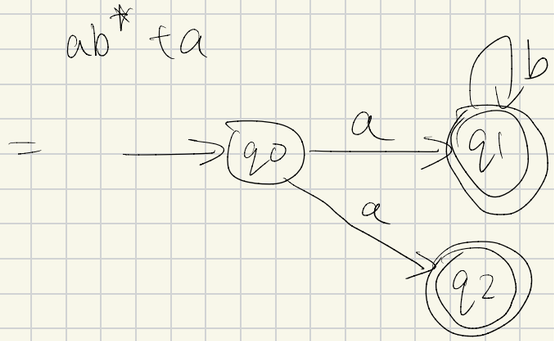
** END **

Question 8 [20%]

Consider the following Regular Expression (RE):
RE = ab* + a

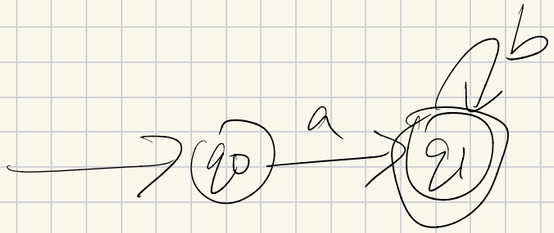
(a) Construct an NFA directly from the given RE without using ε-transitions. [10%]
(b) Convert the directly constructed NFA into an equivalent DFA. [10%]

$ab^* + a$

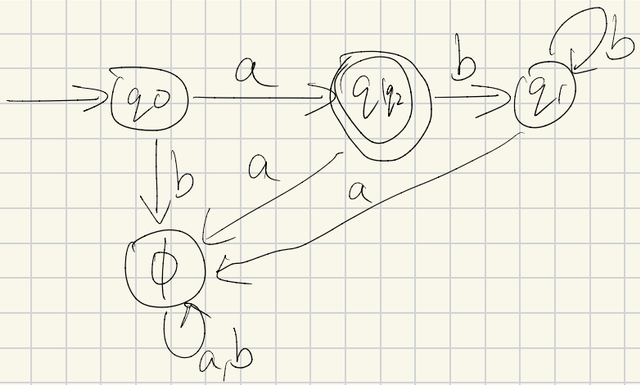


	a	b
q0	q1,q2	—
q1	—	q1
q2	—	—

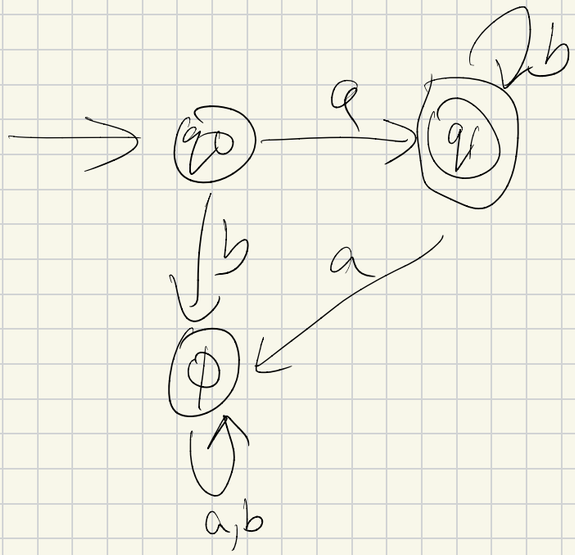
$ab^* + a$



	a	b
q0	q1	—
q1	—	q1



	a	b
q0	q1,q2	q1
q1	Φ	q1
q1q2	Φ	q1
Φ	Φ	Φ



Question 7 [10%]

Consider the following grammar for a section of a language that handles variable assignments and arithmetic operations:

1. S → V = E | V = N
2. E → V + N | V - N
3. V → id | id [N]
4. N → num

In this grammar:
Non-Terminals: V (Variable), E (Expression), N (Number)
Terminals: =, +, -, id, [,], num
Perform left factoring on this grammar.

$S \rightarrow VS'$

$S' \rightarrow = E \mid = N$

$S' \rightarrow = S''$

$S'' \rightarrow E \mid N$

$E \rightarrow VE'$

$E' \rightarrow +N \mid -N$

$E' \rightarrow NE''$

$E'' \rightarrow + \mid -$

$V \rightarrow id\, V'$

$V' \rightarrow \varepsilon \mid [N]$